



University of Niagara Falls – Canada

DAMO 640-08: Machine Learning (Winter Semester 2026)

Assignment 1: Analytical Reasoning in Machine Learning

Group Member:

Nadera Ferguson

Grace Dayrit

Bruno Sassi

Tiago Ceolato

Dr. Touraj BaniRostam

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Introduction

This report analyzes the Clinical risk dataset and reasons about modeling decisions, trade-offs, and limitations. We follow assignment constraints: two base models (Logistic Regression and Decision Tree), one fixed train/test split, at most one hyperparameter per model, and optional PCA with justification. Metrics reported: Accuracy, Precision, Recall, and F1. We believe that these constraints will allow us to compare methods beyond surface-level metrics, and articulate limitations and trade-offs clearly.

Part 1 — Understanding the Dataset

The dataset contains 500 rows and 11 columns (10 predictors + 1 binary target). There are no missing values. The target variable is `risk_status` {0,1} with observed counts: class 0 = 325 (~65%), class 1 = 175 (~35%). Additionally, `visit_year` ranges from 2010 to 2020. Two cholesterol features (`chol_index` and `chol_index_2`) are highly collinear ($r = 0.986$). We aim to answer the question “Is the problem linearly separable?” Not linearly separable because weak pairwise relationships and clinical interaction. In clinical screening, risk often emerges from combinations of factors and non-linear relations. We therefore anticipate that models able to capture interactions or non-linear splits (e.g., Decision Trees) may recover structure that a pure linear boundary cannot (cf. overviews in Microsoft Learn and Google ML Crash Course). (Msakande, n.d.; Google for Developers, n.d.)

Considering the EDA did not reveal any obvious predictors, Welch’s t-tests were performed to measure the statistical significance of the relations. Indicated no statistically significant differences in the mean values of any continuous predictors between individuals with and without `risk_status` (all $p > 0.05$). The smallest observed p-value was for `lifestyle_score` ($p = 0.078$), suggesting weak but non-significant evidence of a group difference. Therefore, considering a contextual qualitative approach, we infer that `age`, `bp_index`, and cholesterol indices are expected to be informative features. And `random_marker_1` and `random_marker_2` are the uncertain feature predictive value and evaluated empirically. We discovered that accuracy alone is insufficient as with a ~65/35 class

distribution, a majority baseline yields ~65% accuracy yet misses all positives. Clinical screening prioritizes reducing false negatives, so Recall is prioritized to minimize missed high-risk patients, while F1 balances precision–recall trade-offs and accuracy provides overall performance context., consistent with standard ML evaluation guidance. (scikit-learn, n.d.-b; scikit-learn, n.d.-c)

Part 2 — Minimal Modeling

We use a single 70/30 train/test split with stratification (random_state=42). Features are standardized for Logistic Regression; Decision Tree uses raw scales. To address strong collinearity between the cholesterol features, our primary configuration drops chol_index_2 (interpretability-first). As a sensitivity analysis, we also evaluate PCA + Logistic Regression. We tune exactly one hyperparameter per model: C for Logistic Regression (set to 1.0) and max_depth for Decision Tree (set to 4). Regularization and dimensionality reduction are standard levers for bias–variance and multicollinearity management. (Google for Developers, n.d.; Encord Team, 2023)

Table 1. Test Metrics

Model	Accuracy	Precision	Recall	F1
Logistic (drop chol_index_2)	0.467	0.000	0.000	0.000
Decision Tree (max_depth=4)	0.580	0.386	0.321	0.351
Logistic + PCA	0.467	0.000	0.000	0.000

Part 3 — Comparative Analysis

Sensitivity to feature representation. Logistic Regression is sensitive to scaling and multicollinearity; coefficients and boundaries shift with redundant features or standardization. Decision Trees are scale-invariant and capture some non-linear interactions via axis-aligned splits but can be unstable with depth changes. (Msakande, n.d.; scikit-learn, n.d.-a). For Logistic Regression, PCA can mitigate instability under strong collinearity (e.g., the cholesterol pair) and may improve generalization, while reducing interpretability because components are latent. For Decision Trees,

PCA rotates the feature space; since trees split on original axes, benefits are limited and interpretability diminishes. (Encord Team, 2023; DataCamp, n.d.)

Logistic Regression can underfit when the boundary is non-linear or interaction-driven (low Recall on positives). Decision Trees with small `max_depth` can underfit (coarse partitions); with large depth, they can overfit spurious patterns. Choosing a single hyperparameter addresses bias–variance without violating constraints. (Google for Developers, n.d.; scikit-learn, n.d.-a)

Clinical cost of errors. False negatives (missed high-risk patients) are usually more costly than false positives. We therefore prioritize Recall and F1, and we also examine threshold adjustments on the positive class probability to trade Precision for higher Recall without changing the model class. (scikit-learn, n.d.-b; scikit-learn, n.d.-c)

Part 4 — PCA as a Decision

With only 10 predictors and a single near-duplicate pair, PCA is not strictly necessary. Our primary path drops `chol_index_2` to retain interpretability. PCA modestly stabilizes the linear model but hinders clinical explainability because components lack direct meaning. An alternative is explicit redundancy removal plus L2-regularized Logistic Regression (via C), or exploring supervised projections (e.g., LDA) in settings where it is permitted. (Encord Team, 2023; GeeksforGeeks, 2023)

Limitations & Next Steps

We restrict ourselves to two simple models and a single split to align with rules. Future work could explore calibration (e.g., Platt scaling), explicit interaction terms for Logistic Regression, cost-sensitive thresholds, and cross-validation for more stable estimates—provided such steps are allowed. (scikit-learn, n.d.-a; scikit-learn, n.d.-b)

AI Tool Use Disclosure

We used an AI assistant to outline the analysis plan, generate boilerplate Python code for t-testing (also for choosing the most adequate test after EDA), data splitting, scaling, optional PCA, Logistic Regression (single hyperparameter C), Decision Tree (single hyperparameter `max_depth`),

and the metrics table. All code was run, validated, and edited by the group on the provided dataset (A1_Clinical_risk.csv). Numerical outputs and interpretations reflect our own execution results.

Conclusion

Through a structured exploration of the Clinical_risk dataset, we found that the problem is not linearly separable and that feature interactions and non-linear patterns play a meaningful role in distinguishing risk outcomes. Our minimal modeling results reinforced this: Logistic Regression, even after addressing collinearity, was unable to identify any positive cases at the default threshold, whereas the Decision Tree captured non-linear structure and achieved materially higher Recall and F1 on the high-risk class. Although PCA provided numerical stability under strong collinearity, it did not improve predictive performance in this setting and reduced interpretability an important consideration in clinical applications. Overall, the Decision Tree offered the most balanced predictive usefulness under the assignment constraints, particularly given the clinical priority of minimizing false negatives. The exercise underscored the importance of understanding data characteristics, evaluating trade-offs in representation, and selecting models that align with real-world costs rather than relying solely on accuracy.

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